Smart Metro Ticket Booking System
IDEATION PHASE

Date	
Team ID	
Project Name	Smart Metro Ticket Booking System
Maximum Marks	2 Marks

1. Problem Statement

1.1 Background

Metro rail is one of the most widely used modes of urban public transport because it is fast, affordable, and reduces road congestion. However, the process of buying a metro ticket in most systems still depends heavily on manual counters, token machines, or partially digital kiosks located inside the station.

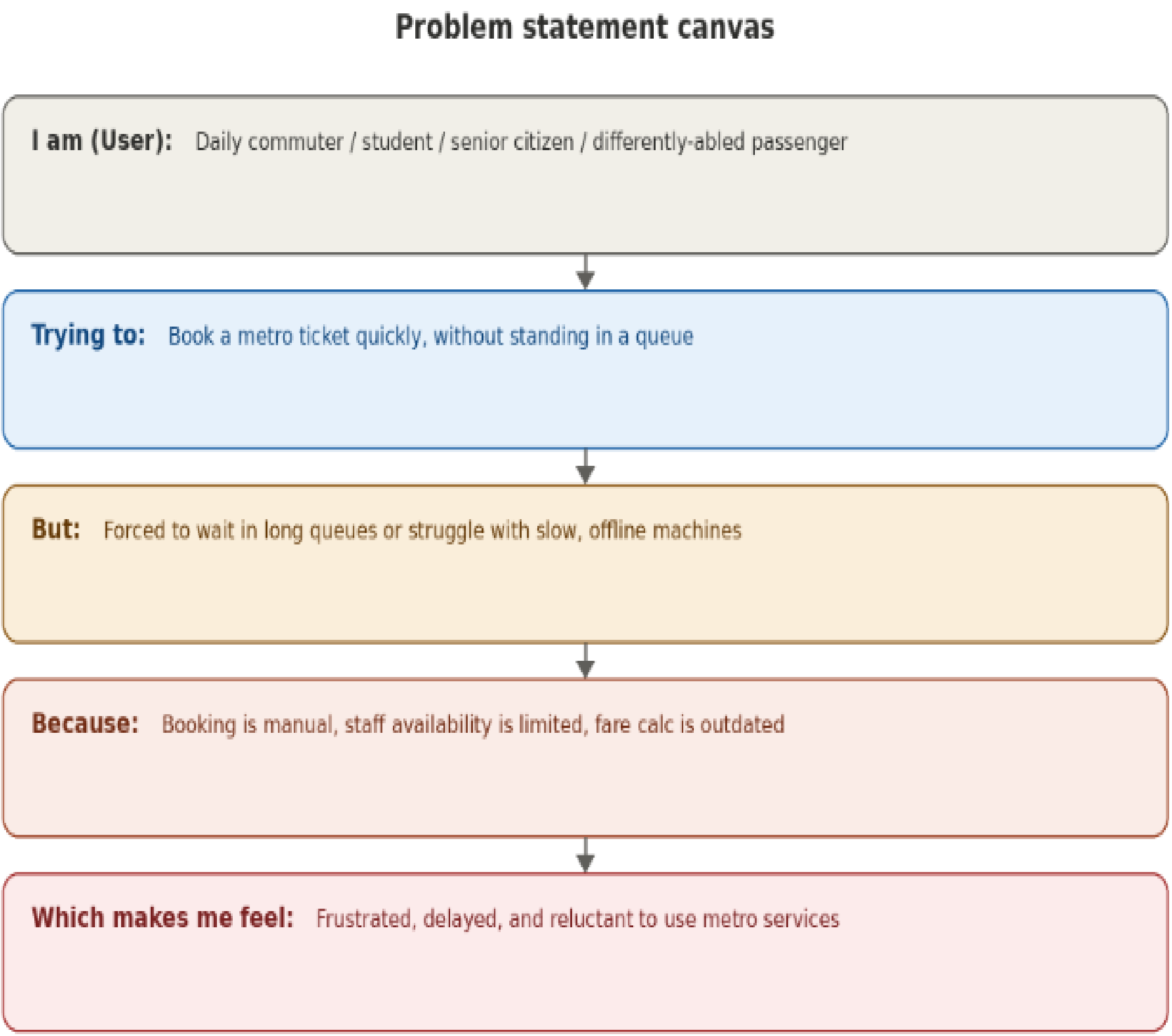
Commuters are required to physically visit the station, stand in a queue, choose the source and destination stations, pay in cash or card, and collect a physical token or paper ticket before they are allowed to enter the platform.

1.2 Problem Statement

"Commuters travelling by metro rail face long queues, manual fare calculation errors, limited counter timings, and dependency on physical tokens/tickets, which leads to delays, overcrowding at ticket counters, and an inconvenient travel experience — especially during peak hours and for first-time or differently-abled passengers."

1.3 Problem Statement Canvas (Who / What / Why)

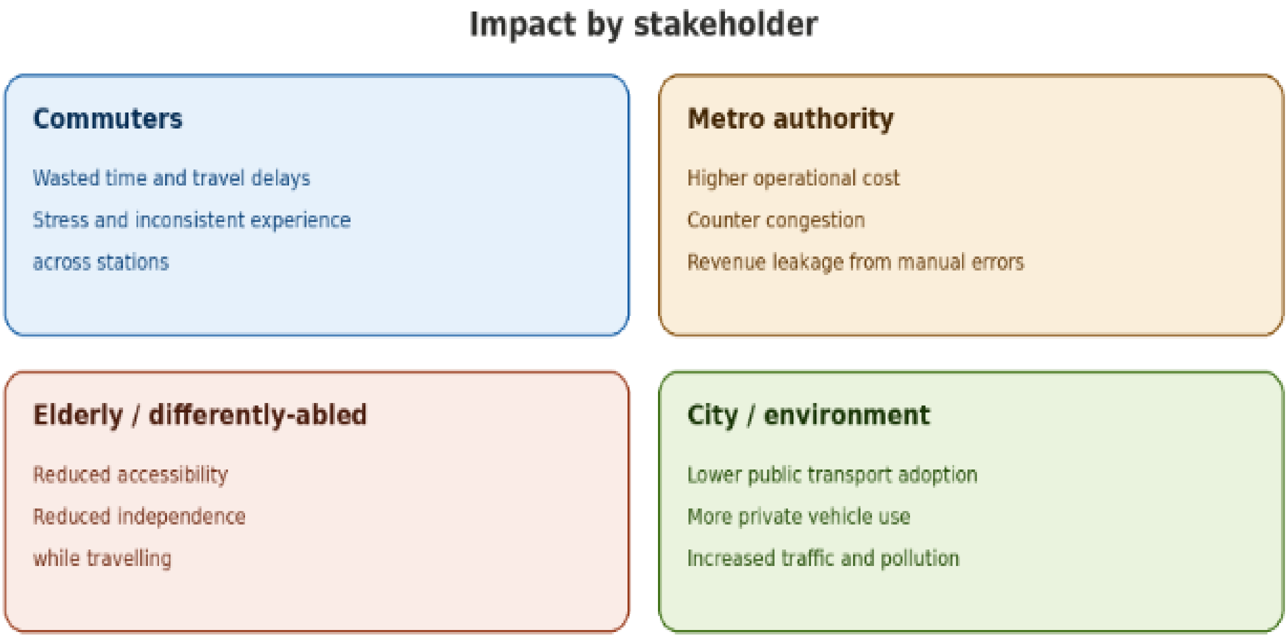
Element	Description
I am (User)	A daily office commuter / student / occasional traveller / senior citizen / differently-abled passenger using the metro.
Trying to	Book a metro ticket quickly for a specific source and destination station without standing in a queue.
But	I am forced to wait in long queues at ticket counters or struggle with vending machines that are slow, offline, or not accessible.
Because	Ticket booking is largely manual, counter staff availability is limited during peak hours, and fare calculation is done manually.



1.4 Key Points

- Long queues at ticket counters during office hours, exams, and festival rush.
- Manual fare calculation is time-consuming and prone to human error.
- Limited operating hours of physical counters versus 24x7 travel needs.
- Loss or damage of physical tokens/paper tickets causing re-verification delays.
- Little to no support for elderly, disabled, or visually impaired commuters at manual counters.
- No unified digital record of past journeys, spending, or trip history for commuters.
- High operational cost for metro authorities in maintaining counters, staff, and token inventory.
- No integration between fare collection and workflow-driven customer support or refunds.

1.5 Impact of the Problem



1.6 Need for a Digital Solution

There is a clear need for a digital, self-service metro ticket booking system that lets commuters select their stations, get an automatically calculated fare, and instantly receive a QR-code based digital ticket — all without visiting a counter. Building this on the ServiceNow platform allows the solution to leverage the Service Catalog for a guided booking experience, Flow Designer for automated fare calculation and ticket generation, and Notifications for instant delivery of the digital ticket, resulting in a faster, more accessible, and more reliable travel experience.